LECTURE 8: MATH1061/7861
TOPICS: DIRECT PROOFS AND COUNTEREXAMPLES
DATE: WEDNESDAY MARCH 11, 2026

Learning objectives:

(1) Learn how to write a direct proof.

(2) Learn how to disprove a statement by exhibiting a counterexample.

* Make sure to keep up with videos and Pre-Lecture Readings. Starting next week things will get more complicated.

Q1: Prove or disprove:

If $x \in \mathbb{Z}$ is such that $x+2 = 4y$ for some $y \in \mathbb{Z}$, then $\frac{x}{2}$ is an odd integer.

We prove the statement as follows.

Suppose $x \in \mathbb{Z}$ and $x+2=4y$ for some $y \in \mathbb{Z}$. Then $x = 4y-2 = 2(2y-1)$.

Therefore, $\frac{x}{2} = 2y-1 = 2(y-1)+1$.

So we see that $\frac{x}{2}$ is odd $\square$

Remark: Proofs should make sense from English language perspective.

Q2: Prove or disprove:

For each integer $n \geq 2$, the product of the first n prime numbers, minus 1, is prime.

Let $P_1, P_2, \dots, P_n$ be the first n prime numbers. Then $P_1 \times P_2 \times \dots \times P_n - 1$ is also a prime number.

(Recall a prime number is an integer > 1 such that its only (integer) factors are one and itself

~~B~~ ~~1~~, 2, 3, ~~4~~, 5, ~~6~~, 7, ~~8~~, ~~9~~, ~~10~~)

Let $n = 4$. Then $P_1 P_2 P_3 P_4 - 1 = 2 \times 3 \times 5 \times 7 - 1$

$= 209 = 11 \times 19$.

So the statement is False.

Q2) Continued.

To get some intuition, we try out the first few n's:

$n=2$: $P_1 P_2 - 1 = 2 \times 3 - 1 = 5$ Prime

$n=3$: $P_1 P_2 P_3 - 1 = 2 \times 3 \times 5 - 1 = 29$ Prime

$n=4$: $P_1 P_2 P_3 P_4 - 1 = 2 \times 3 \times 5 \times 7 - 1 = 209$

209 is not a prime. It is divisible by 11.

$209 = 11 \times 19$.

$$\mathbb{N} = \{0, 1, 2, 3, \ldots\}$$

Q3 ~~Q3~~: Prove or disprove:

$$\exists n \in \mathbb{N} \text{ s.t. } n^2 + 5n + 6 \text{ is prime.}$$

We claim the statement is false, i.e.

$\forall n \in \mathbb{N}$, n^2+5n+6 is composite.

We offer two proofs.

Proof 1: Note that

$$n^2+5n+6 = (n+2)(n+3)$$

Moreover, $n+2 > 1$, $n+3 > 1$.

So n^2+5n+6 is always composite.

Proof 2: We claim that n^2+5n+6 is always even. Since it is automatically >2, it follows that it must be composite.

Indeed, if $n = 2K \Rightarrow n^2+5n+6 = 4K^2+10K+6$

$$= 2(2K^2+5K+3)$$

if $n = 2K+1 \Rightarrow n^2+5n+6 = (2K+1)^2 + 5(2K+1)+6$

$$= 4K^2+4K+1+10K+5+6$$

$$= 4K^2+14K+12 = 2(2K^2+7K+6)$$

Q4

~~Q3~~: Prove or disprove:

For all $n \in \mathbb{Z}$, if n is odd then $3n + 2$ is odd.

~~Counter-example:~~

~~Consider $n=1$. Then n is odd~~

~~but $3n+2 = 3 \times 1 + 2 = 6$ is even.~~

~~Thus the statement is false.~~

Proof:

Take any odd integer n, so $n = 2k+1$ for some $k \in \mathbb{Z}$. Then

$$3n + 2 = 3(2k+1) + 2 = 6k + 5 = 2(3k+2) + 1.$$

As $3k+2$ is an integer, $3n+2$ is odd. $\square$